AIM 1048: review note

Can neighboring water-wave Bloch bands overlap?

5 November 2026

Claim status: solution claimed, awaiting independent review.

Submitted by Siavash Sadeghi. The proof and audit records were produced with AI assistance. No human has reviewed this work, and no formal verification is claimed. The submission proposes the repository status *Solution claimed*, pending further independent mathematical review.

1. Exact target and covered class

The repository is https://github.com/MColbrook/AIM. This package addresses problem 1048 at revision

`4f5b1659355cb61d194da94ef14615214e1363a8`.

The exact problem page and contribution instructions at that revision are preserved in `TARGET_SNAPSHOT.md` and `CONTRIBUTING_SNAPSHOT.md`. The repository status at that revision is *Partially resolved*. The proposed update preserves the original problem statement and its ID.

Fix $h > 0$ and a real, 2π-periodic $b \in C^1(\mathbb{R})$ with $d_- := \min_x(h - b(x)) > 0$. In

$$\Omega_b = \{(x, y) : -h + b(x) < y < 0\},$$

extend top traces harmonically, with homogeneous Neumann data on the bottom, finite energy per period cell, and character $e^{2\pi i\theta}$. The free-surface Dirichlet–Neumann operator is $f \mapsto \partial_y \Phi(x, 0)$ in physical $L^2(dx)$. Order the eigenvalues separately in each fiber, including multiplicities: $\Lambda_0(\theta) \le \Lambda_1(\theta) \le \cdots$.

The proposed theorem is

$$\boxed{\Lambda_n(\theta_a) \le \Lambda_{n+1}(\theta_b) \quad (n \ge 0,\ \theta_a, \theta_b \in \mathbb{R}).}$$

Phase continuity and compactness of $[-\frac{1}{2}, \frac{1}{2}]$ give precisely the repository target

$$\max_\theta \Lambda_n(\theta) \le \min_\theta \Lambda_{n+1}(\theta).$$

Conversely the max–min target implies each displayed pairwise inequality, so these formulations have the same strength.

There is no additional restriction on the profile. The proof claims the whole stated C^1 graph class, not just analytic or small profiles. The bottom may be non-even and have nonzero mean; its amplitude and slope have no imposed smallness bound. The given period need not be fundamental. Every fixed profile has finite depth bounds and bounded slope by periodicity and C^1 regularity, but no uniform bound across profiles is assumed. Band touching and eigenvalue multiplicity are allowed. The common form domain is $H^{1/2}$; its closed realization is the natural extension of the harmonic problem stated for H^1_{loc} traces.

2. Map of the proof and hypothesis comparison

The full argument is in `research_nonoverlap_proof.tex` and its PDF. Its six sections have the following roles.

Sections 1 and 5: the exact geometry supplies the mass identity. For an analytic graph, a conformal map Z takes a strip of depth $d > 0$ and horizontal period $P = 2\pi$ to the physical channel, with $Z(w + P) = Z(w) + P$. It preserves the Bloch character and Dirichlet energy. The top measure is $dx = \rho(s)ds$ with $\rho(s) = Z'(s) > 0$. The bottom horizontal derivative is

$$p(s) = \operatorname{Re} Z'(s - id) > 0, \qquad p_k = \cosh(dk)\rho_k.$$

Positivity of p follows from the regular graph geometry. It is not inferred from positivity of an arbitrary surface weight ρ.

For a holomorphic representation V on the doubled strip, put $V_\pm(s) = V(s \pm id)$ and $V_0(s) = V(s)$. The energy and physical mass are

$$Q(V) = \frac{1}{2} \int_{(0,P)\times(-d,d)} |V'|^2,$$
$$M(V) = \frac{1}{4} \int_0^P \rho(|V_+|^2 + |V_-|^2) + \frac{1}{2} \int_0^P p|V_0|^2.$$

The cross term uses $\operatorname{Re} \int \rho V_+ \overline{V_-} = \int p|V_0|^2$. Taking the real part is essential; the identity permits complex Bloch data and non-even weights.

Sections 2 and 3: one zero changes any nontrivial character. For $-d < y < d$, a normally convergent strip Blaschke product B_y has one simple zero per period, unit modulus on both horizontal edges, and character parameter $\chi = (1 - y/d)/2 \in (0, 1)$. Thus all nontrivial characters occur. If V vanishes at its chosen zero, $W = V/B_y$ is holomorphic and represents a trace at phase $\theta - \chi$. On the smooth functions used in min–max,

$$M(W) \geq M(V), \qquad Q(V) - Q(W) = \frac{1}{4} \sum_{\pm} \int_0^P (\partial_\nu \log |B_y|)\, |W_\pm|^2 \geq 0.$$

The quotient singularities are removable. Lateral boundary terms cancel because both characters have unit modulus.

Section 4: ordered min–max for arbitrary phase pairs. Approximate a source spectral space of complex dimension $n + 2$ at phase θ_b by finite Fourier traces. A single complex evaluation constraint leaves dimension at least $n + 1$. Choose $\chi = \theta_b - \theta_a$ modulo one. Division and top reconstruction are injective and do not increase the Rayleigh quotient. Min–max gives the claimed ordered comparison. Coincident characters use ordinary eigenvalue ordering. This step assumes neither simplicity nor tracking of analytic eigenvalue branches.

Section 6: all C^1 graphs. Approximate b in C^1 by real trigonometric polynomials, retaining a positive lower depth. Pulling bulk energies back by $y = (h - b(x))z$ gives uniformly convergent, uniformly elliptic coefficient matrices. Relative form comparison yields convergence of every fixed ordered eigenvalue. Pass the inequality for each fixed index and phase pair. A separate bulk estimate proves phase continuity and hence existence of the band extrema.

3. Review provenance and questions for a reviewer

The user requested an adversarial audit of the Blaschke factor, the mass identity and the all-phase min–max argument. Three fresh AI review contexts read the proof without the earlier audit reports. The coordinating AI also checked the central identities and numerical diagnostics.

`fresh_blaschke_audit.txt` Product convergence and nonvanishing, exact zero set, boundary regularity, character sign, quotient removal and energy comparison. Its stated scope does not independently cover the geometric reduction and C^1 limit.

`fresh_mass_audit.txt` Complex reconstruction, the real-part mass identity, conformal measure and positivity forced by graph geometry.

`fresh_minmax_audit.txt` Complex dimension, arbitrary phase differences, ordered indices, multiplicities, finite Fourier approximation and the C^1 passage.

`research_nonoverlap_fresh_audit.txt` Coordinating audit and the combined findings, including stress-test limitations.

These reports found no fatal flaw within their stated scopes. Their terminology "independent reviewer" refers to separate AI contexts in the same workflow. It does not mean independent human peer review. The reports prompted expository clarifications now present in the proof; they are not cryptographic attestations of the final file revision. No human review has been reported. The package is a proposed mathematical argument for examination, not a certification of correctness by the repository or by a formal proof assistant.

The main points for fresh scrutiny are:

1. Does the normalized product converge to a nonzero holomorphic function away from exactly the prescribed zeros, with the stated character and sufficient regularity through both edges?
2. Does the symmetrized cross term equal the middle-line mass for all complex Bloch data, and is $p > 0$ supplied by every admissible analytic graph?
3. Does one complex constraint, followed by injective reconstruction, give the correct $n + 2$ to $n + 1$ dimension comparison for every phase pair?
4. Does physical-domain form convergence remove the analytic-profile restriction without adding a uniform slope, amplitude or spectral-index assumption?

Numerical evidence is supplementary. The included NumPy script checks 21 cases with complex coefficients, non-even positive weights, several strip depths and phases near the character endpoints. It checks product, phase, mass and energy identities and records reconstruction errors. A deep-strip case has a large top-to-bottom norm ratio and visible relative cancellation at the smaller bottom trace; the report retains that limitation. The tests are ordinary floating-point diagnostics, not interval certificates, exhaustive counterexample searches or proof dependencies. A successful run cannot establish the theorem.

4. Reproduction, exclusions and references

Both TeX sources are standalone articles. From the extracted package directory, compile each twice with `pdflatex`, using `-no-shell-escape -interaction=nonstopmode -halt-on-error`. The required packages are standard: `amsmath`, `amssymb`, `amsthm`, `geometry` and `hyperref`. No bibliography database, custom style, figure or additional TeX input is required. The README gives complete commands and optional diagnostic instructions.

`metadata.json` pins the repository revision and records source hashes and provenance. `MANIFEST.json` gives packaged-file hashes and sizes. `BUILD_VERIFICATION.txt` records the checks actually performed during packaging. The mathematical text of the proof is unchanged by packaging; only attribution, a review-status notice and line wrapping are edited. The four supplied AI audit reports and recorded diagnostic results are preserved byte for byte. Diagnostic validation now uses explicit exceptions so that Python optimization cannot disable it. A separate submission review records the final source hashes and the checks reproduced for this PR.

Scope boundaries. This claim concerns finite-depth, flat-top, two-dimensional graph channels with homogeneous Neumann bottom data. It does not assert the theorem for arbitrary positive strip weights, nongraph channels, obstacles, pinched depths, different boundary conditions or a shallow-water approximation. It also does not claim endpoint localization of all band edges, monotonicity of band functions, simplicity or strict opening of every gap. None of those stronger statements is part of problem 1048's target.

The cited mathematical background and primary problem sources are:

1. L. V. Ahlfors, *Complex Analysis*, third edition, McGraw–Hill, 1979. Chapter VI, §5.1, Theorem 10, pp. 255–256: annular uniformization in connectivity two. Chapter VI, §§1.3–1.4, Theorems 3–4, pp. 233–235: the local reflection argument used after straightening analytic boundary arcs. The proof checks the separated, analytic embedded boundary components needed here.

2. T. Kato, *Perturbation Theory for Linear Operators*, second edition, corrected printing, Springer, 1980. Chapter VI, Theorem 2.1: representation of a closed form by its associated operator. The proof states the form domain and compact embedding and supplies the required min–max and phase comparisons.

3. W. Craig, M. Gazeau, C. Lacave and C. Sulem, "Bloch theory and spectral gaps for linearized water waves," *SIAM J. Math. Anal.* **50**(5) (2018), 5477–5501. DOI: 10.1137/17M113561X. The nonoverlap question is stated separately from band-edge localization; see the introduction after Definition 1.1.

4. C. Lacave, M. Ménard and C. Sulem, "Bloch–Floquet band gaps for water waves over a periodic bottom," *EMS Surv. Math. Sci.* **12**(1) (2025), 243–288. DOI: 10.4171/EMSS/93. The repository cites §4, p. 270 for the general ordering question and §3.4 for ordered eigenvalues versus analytic relabelings.

The problem snapshot preserves the repository's dated literature review. This package makes no claim of priority or exhaustive knowledge of all subsequent work. Original books and papers are not redistributed here. See the package README and resolution report for the submission links, current reproduction record and explicit review limitations.